

Sampling What We Cannot Sum

Markov Chain Monte Carlo at the Frontier

Lecture 4's **walker**, a Markov chain, visits each candidate answer in proportion to its probability, **without the impossible sum** Z . It is behind black-hole measurements, image generators and a new way to sample language models. Today you see it at the frontier, and try it yourself.

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Builds on Lecture 4: Sampling replaces an impossible sum

Outline

► Sampling without Z (Lecture 4)

Troubles 1–3: measuring unknowns from data

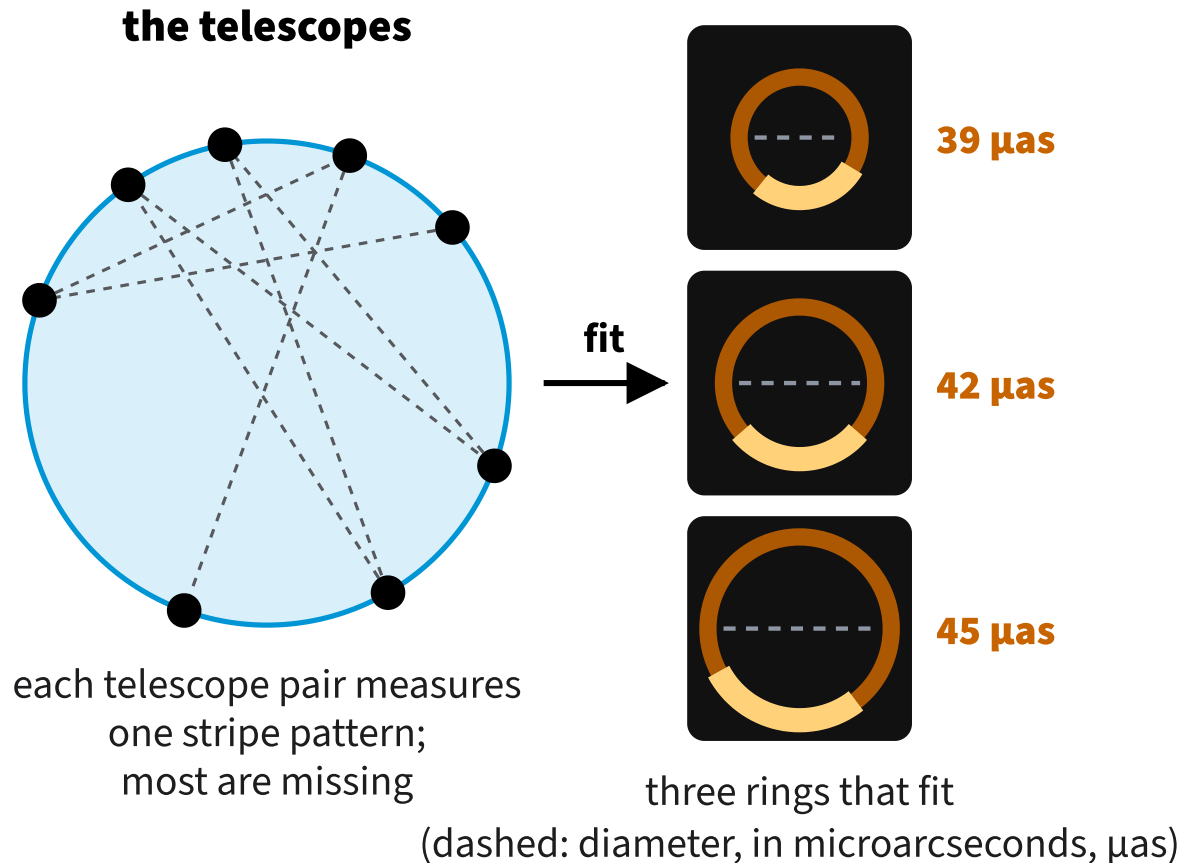
Troubles 4–5: creating pictures and text

Conclusion and your turn

One story: the sampler of Lecture 4, which we call **the walker**, meets five troubles at the frontier, and one small change fixes each. First: why sample at all?

Sampling without Z

April 2019: the first picture of a black hole.



Look: telescopes across the Earth gave only a few measurements, and all three rings on the left fit them.

Q: Which picture is right?

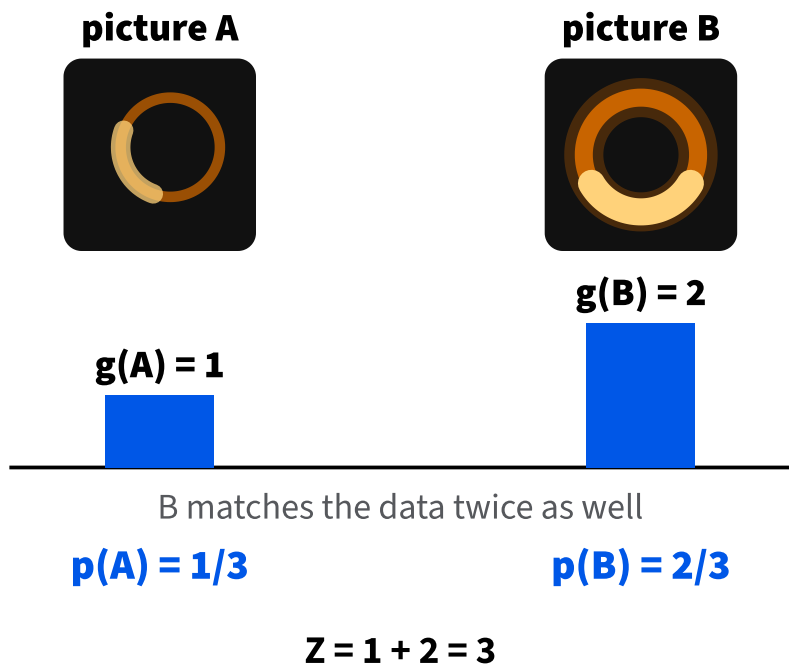
A: Many fit. Pick among them at random, better fits more often: their spread is the error bar.

Sampling without Z : in three sentences

- BACKGROUND** Data rarely pin down one answer: many answers fit, some better than others.
- KEY IDEA** Pick answers at random, better fits more often. Lecture 4's **walker** does this by comparing just two answers at a time, so it never needs the **impossible** total Z .
- USED FOR** Answers with error bars, from the size of a black hole to the effect of a medicine.

Sampling without Z

Main idea: **sample**, that is, pick a picture at random, with probability proportional to its **fit score**.



$$p(x) = \frac{g(x)}{Z}$$

← fit score: easy

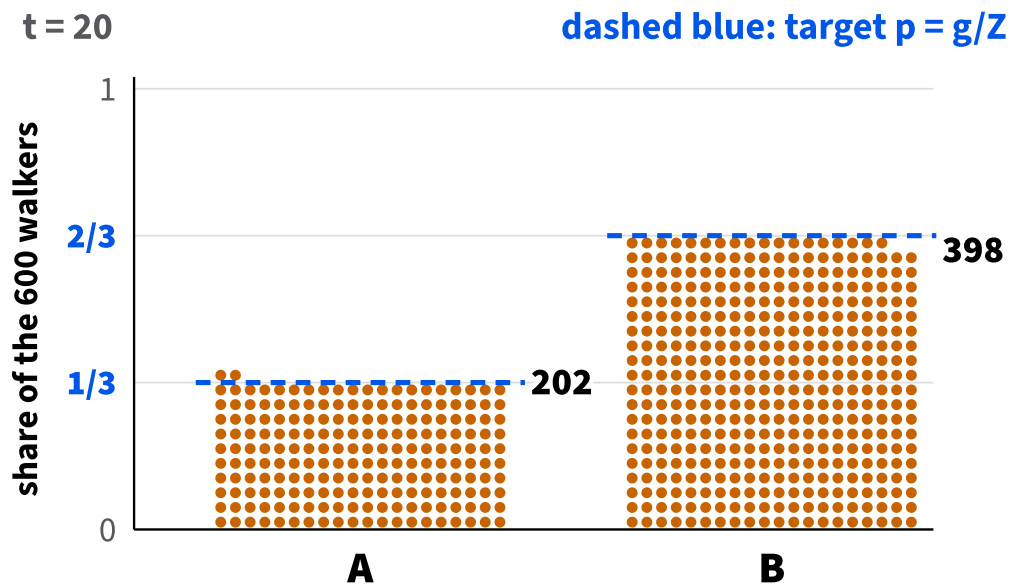
← sum of all fit scores: hard

Example Fit scores 1 and 2, so pick A one third and B two thirds of the time.

Look: one fit score is easy to compute. Here $Z = 1 + 2 = 3$, but for a real image the sum has more terms than there are atoms in the universe.

Sampling without Z

The trick: compare two pictures, and Z cancels.



$$\frac{p(\text{new})}{p(\text{old})} = \frac{g(\text{new})/Z}{g(\text{old})/Z} = \frac{g(\text{new})}{g(\text{old})}$$

The walker (a **Markov chain**) holds one picture. Each step it proposes another, and **the accept test** decides: move or stay [Metropolis et al., 1953; Hastings, 1970].

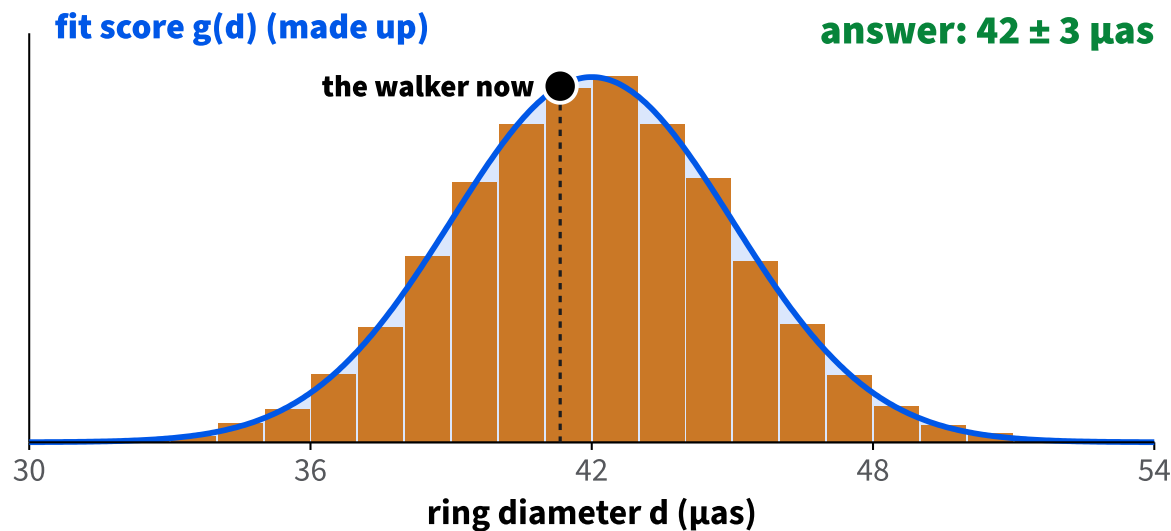
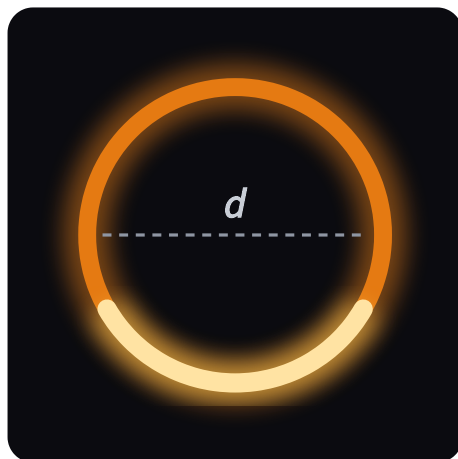
- ▶ **Better fit** ($A \rightarrow B$): always move.
- ▶ **Worse fit** ($B \rightarrow A$): move with probability $g(\text{new})/g(\text{old}) = \frac{1}{2}$.

Watch: 600 **walkers** start at A. Their **visits**, the places they have been, settle near one third at A and two thirds at B (**p. 35**). These visits are the **samples**.

Try: set the **fit score** of B to 3. Where will they settle?

Sampling without Z

One **walker** hops between ring sizes; where it spends its time gives **the answer**.



Watch: each step proposes a nearby size and applies the **accept test** (p. 6); the ring shows the current size, and the orange bars count the **visits**. Their centre is the answer, their width the **error bar**. **Try:** tiny or huge steps.

Q: Why not just take the best-fitting ring? A: It has no error bar, and nearby rings fit almost as well.

Result: 42 ± 3 microarcseconds (μas), about an orange on the Moon seen from Earth; mass 6.5 ± 0.7 billion Suns [Event Horizon Telescope Collaboration, 2019].

Main Question

**The walker needs only ratios of fit scores, never Z .
Can it also weigh colliding black holes, measure how
much COVID-19 lockdowns helped, give neural
networks error bars, paint pictures, and help language
models reason?**

This is indeed the case, after a few small changes.

Keep this question in mind: each part changes the same walker in one small way.

Sampling without Z

Six words for the rest of the lab.

Word	What it means today	Name in Lecture 4
candidate x	one possible answer: a picture, a ring size, a set of network weights, a sentence	a state
fit score $g(x)$	how well x matches the data; Z is the sum of all fit scores, and $p = g/Z$	weight (energy $E = -\ln g$)
the walker	one point that jumps between candidates: propose, then the accept test	Markov chain (Metropolis–Hastings)
accept test	keep a proposed jump with probability $\min\{1, g(\text{new})/g(\text{old})\}$ (p. 6)	acceptance step
visits	the candidates the walker has been at; their histogram approaches p	samples
hill, valley	a region of good fits; the bad fits between them. Lecture 4 drew the energy $-\ln g$, so its valleys are our hills.	mode, barrier

Read: so “the **visits** fill a hill” means “the **samples** cover a region of good answers”.

Sampling without Z

Five troubles at the frontier, five fixes.

	Trouble	Fix	Used for
1	far-apart hills	???	colliding black holes
2	many unknowns at once	???	COVID-19 lockdowns
3	too much data	???	neural-network error bars
4	no formula for g , only examples	???	image generators
5	sharpening: a new Z over all texts	???	language-model reasoning

Read: troubles 1–3 appear when the **walker** measures unknowns, 4–5 when it creates. The fixes are revealed one by one.

Outline

Sampling without Z (Lecture 4)

► **Troubles 1–3: measuring unknowns from data**

Troubles 4–5: creating pictures and text

Conclusion and your turn

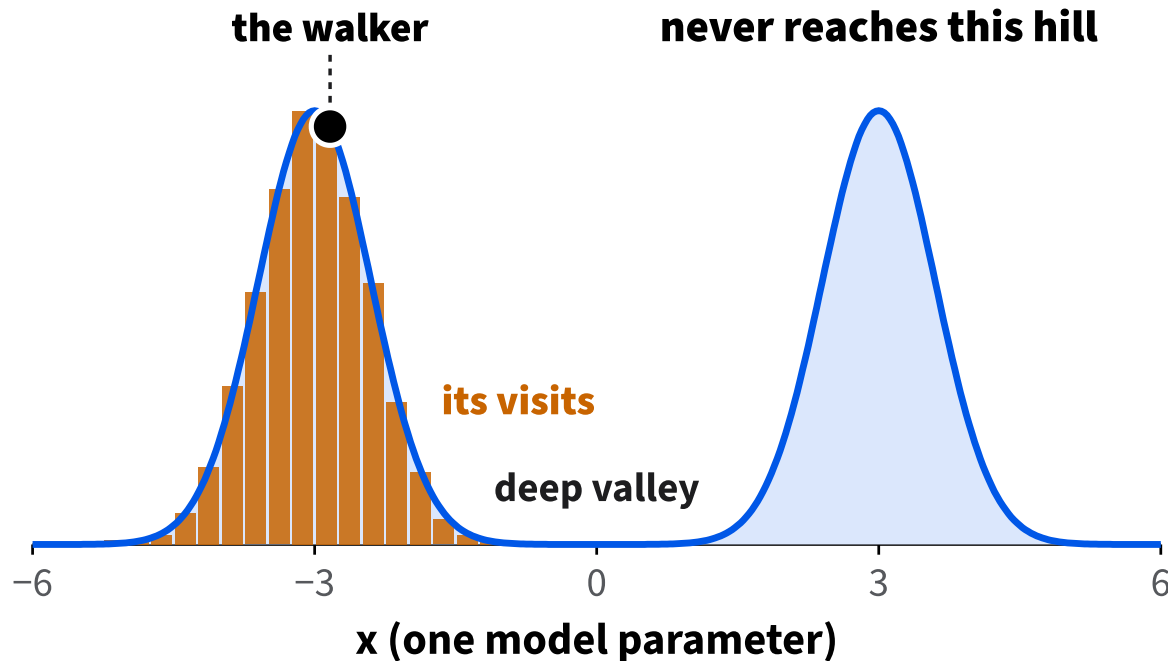
Next: three ways the walker fails when it measures, and the small change that fixes each.

Trouble 1 in three sentences

- BACKGROUND** Sometimes two very different answers both fit well (**two hills**), with bad answers in between.
- KEY IDEA** Run a second **walker** on **flattened fit scores**, so it crosses easily, and let the two **walkers swap** places now and then.
- USED FOR** Weighing two colliding black holes from a gravitational wave (LIGO, 2015), a discovery that won the 2017 Nobel Prize in Physics.

Trouble 1: far-apart hills

Trouble: two far-apart hills trap the walker.



Watch: its **visits** fill only the left hill. The right hill, an equally good group of answers, is **never found**, and nothing warns us.

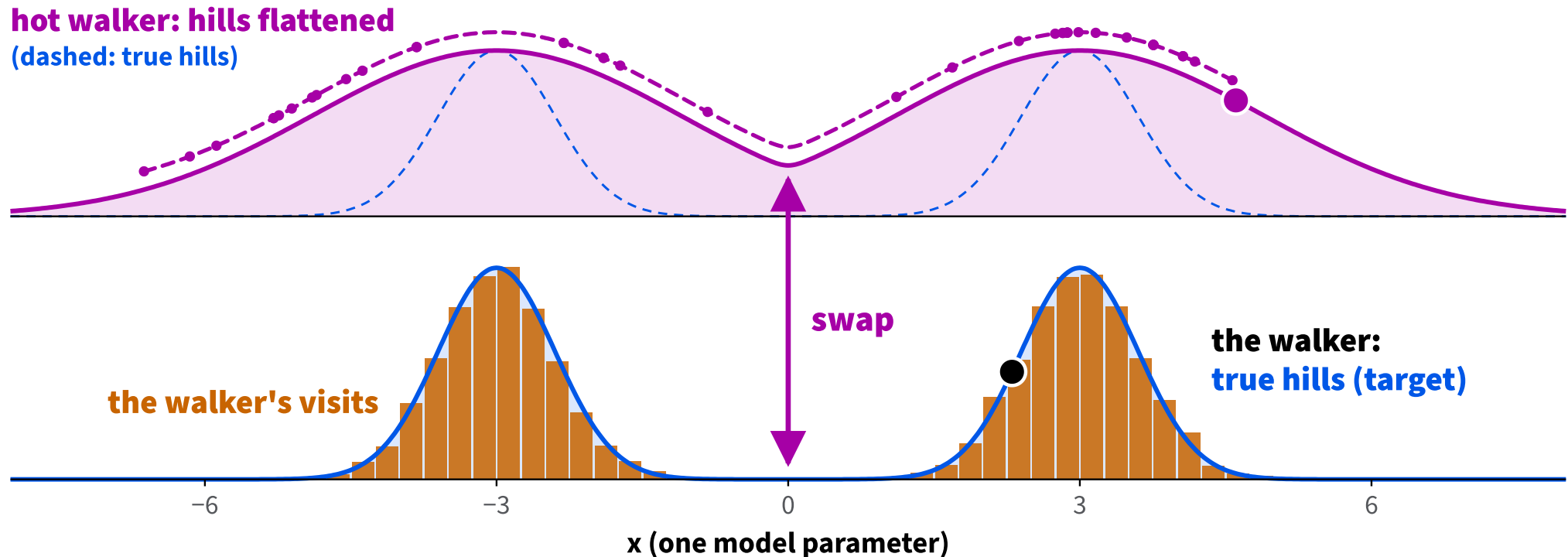
Q: Why is it stuck?

A: Crossing needs many worse-fit moves in a row, each rarely accepted.

Try: bring the hills closer.

Trouble 1: far-apart hills

Fix: run a second, **hot walker** on **flattened** hills; now and then the two **swap** places.

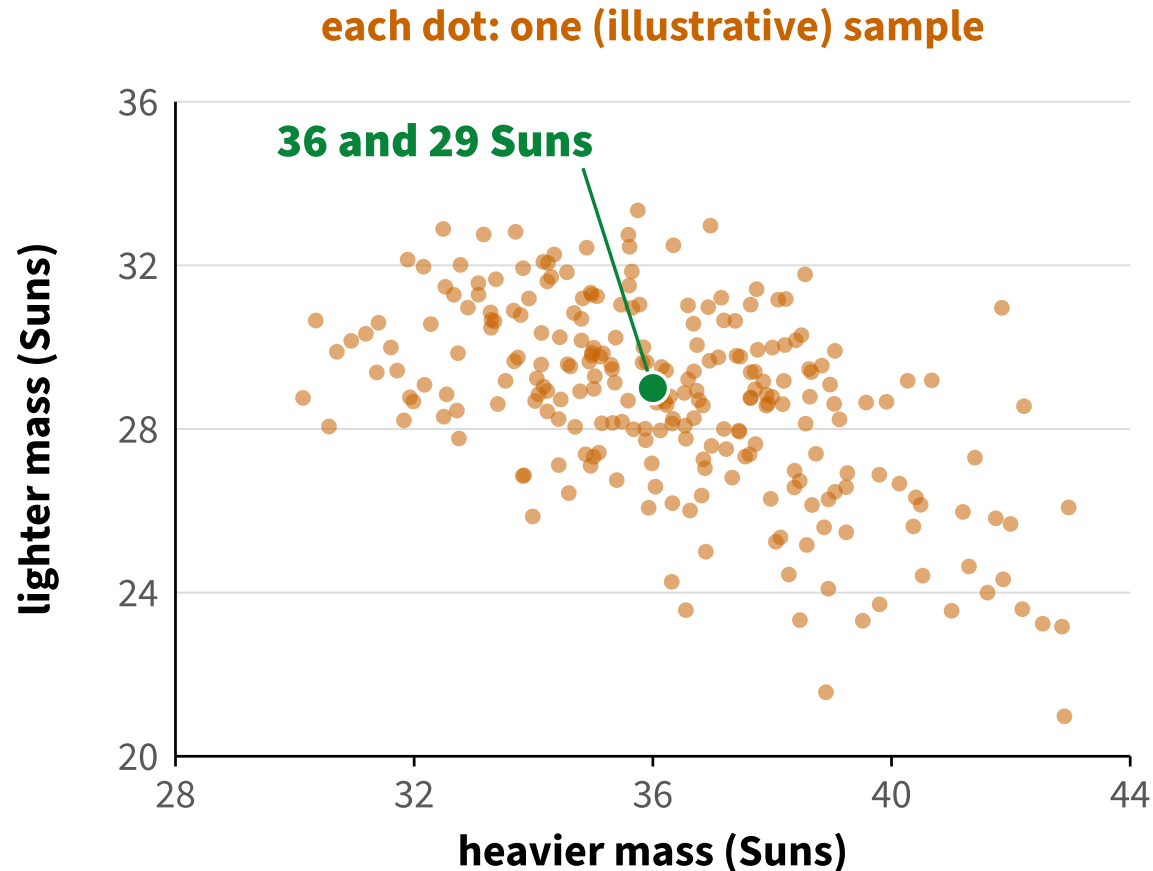


Watch: the hot walker (top) uses the **fit score** to the power $1/T$, which flattens the hills, so it crosses. At each magenta swap the two exchange positions after their own **accept test** (p. 37), so the walker's **visits** fill both hills.

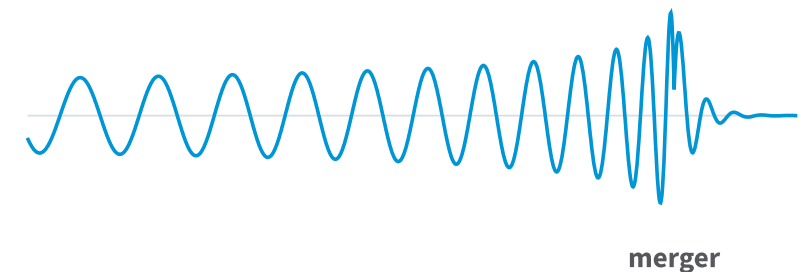
Try: $T = 1$. Does the hot walker still help? This is **parallel tempering** [Swendsen and Wang, 1986; Geyer, 1991].

Trouble 1: far-apart hills

Result: two colliding black holes were weighed with hot and cold **walkers**.



the measured wave (schematic)



14 September 2015: the Laser Interferometer Gravitational-Wave Observatory (LIGO) caught a 0.2-second chirp, a rising whistle, from two black holes merging 1.3 billion light-years away [Veitch et al., 2015; Abbott et al., 2016]. It won the 2017 Nobel Prize in Physics.

Look: each dot is one visit (a **sample**) of the two masses; the cloud's centre is the estimate, its size the **error bar**.

Trouble 2 in three sentences

BACKGROUND

Real models have hundreds of unknowns, and then a random step almost always lands on a bad answer.

KEY IDEA

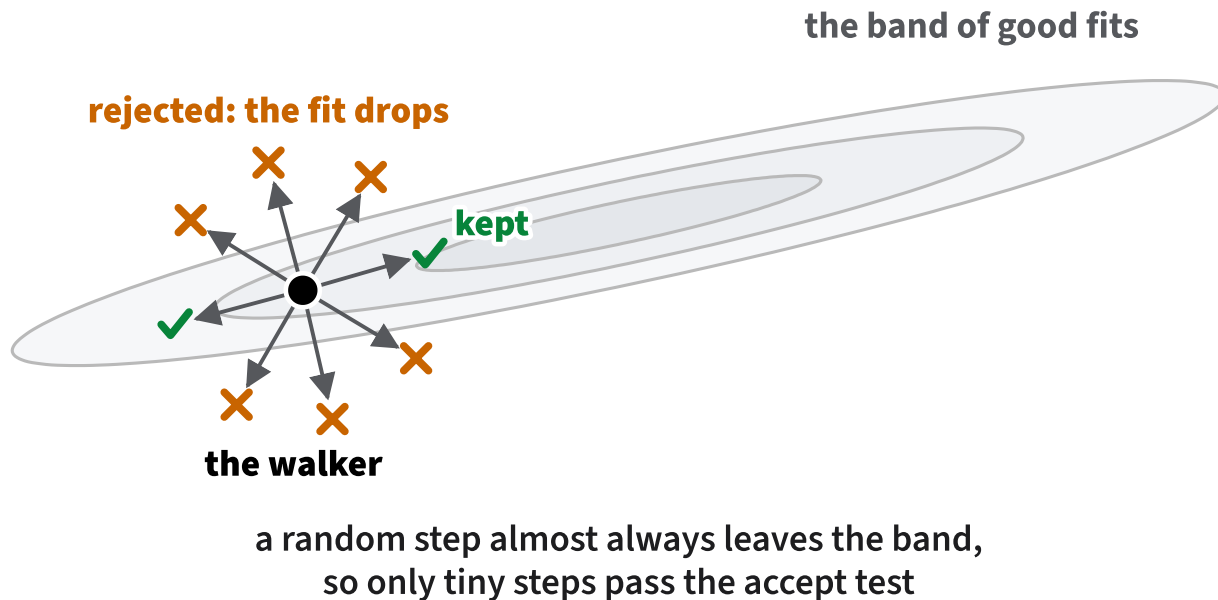
Give the **walker** a push and let it **slide like a marble** along the good answers, steered by the **slope** of the **fit score**.

USED FOR

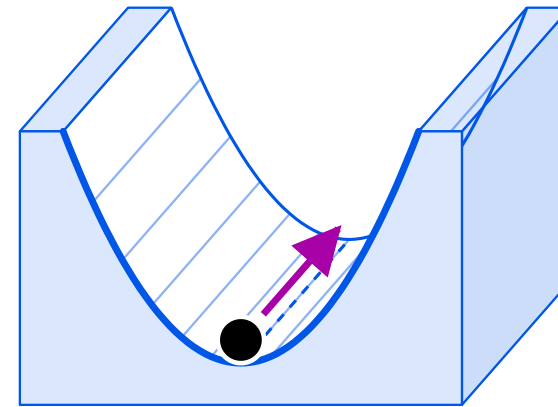
The Stan software, used to measure how much COVID-19 lockdowns helped in 11 countries (2020).

Trouble 2: many unknowns at once

Trouble: with many unknowns, good fits form a **narrow band**, and random steps leave it.



flip the fit upside down:
good fits = bottom of a groove
blue: the height $-\ln g$ across the band

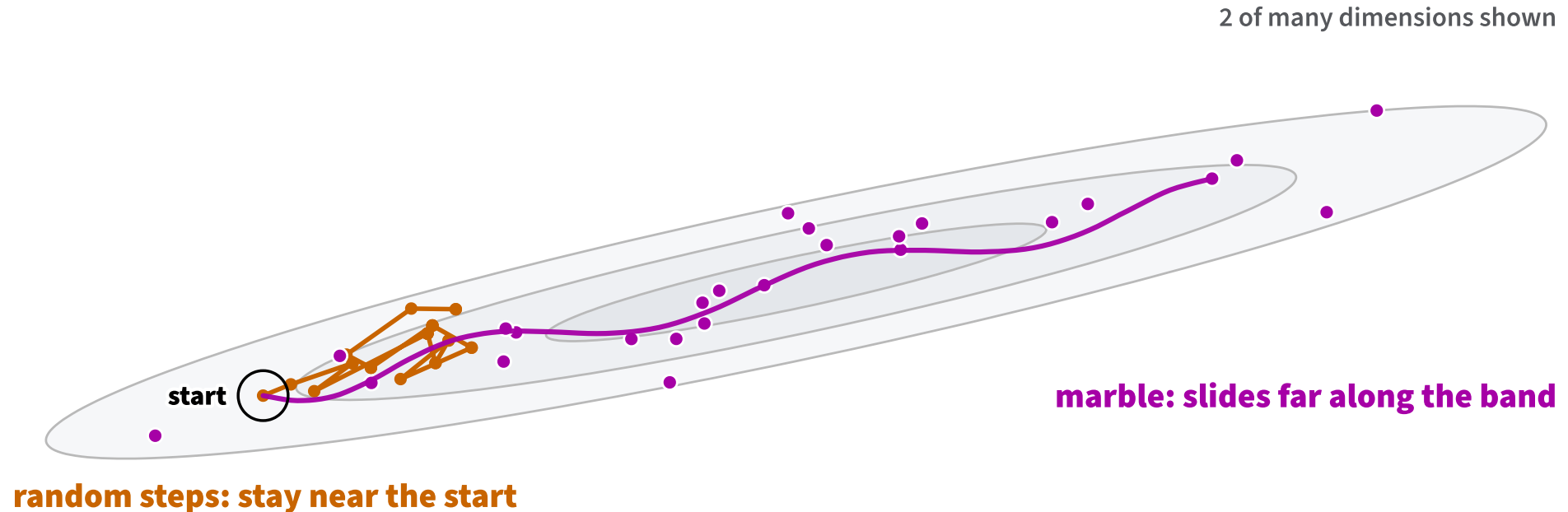


a pushed marble **slides along the groove**,
not out of it

Look: almost every random step leaves the band (×) and is rejected; only tiny steps are kept, so the **walker crawls**. On the right, the **fit score** is flipped upside down: the band becomes the bottom of a **groove**.

Trouble 2: many unknowns at once

Fix: push the **walker** and let it slide along the **groove**, like a **marble**.



grey: contour lines of the target · dots: where each roll ended

Watch: in 25 moves the random walker (orange) stays near the start, while the marble (magenta) slides far along the band. It keeps its energy, height plus $\text{speed}^2/2$, so even far moves pass the **accept test**. **Try:** roll length 1.

This is **Hamiltonian Monte Carlo** [Duane et al., 1987]; the Stan software used it to estimate how much lockdowns slowed COVID-19 in 11 European countries [Flaxman et al., 2020].

Trouble 3 in three sentences

BACKGROUND

For a neural network, one **fit score** needs all the training data, **millions** of examples, at every step.

KEY IDEA

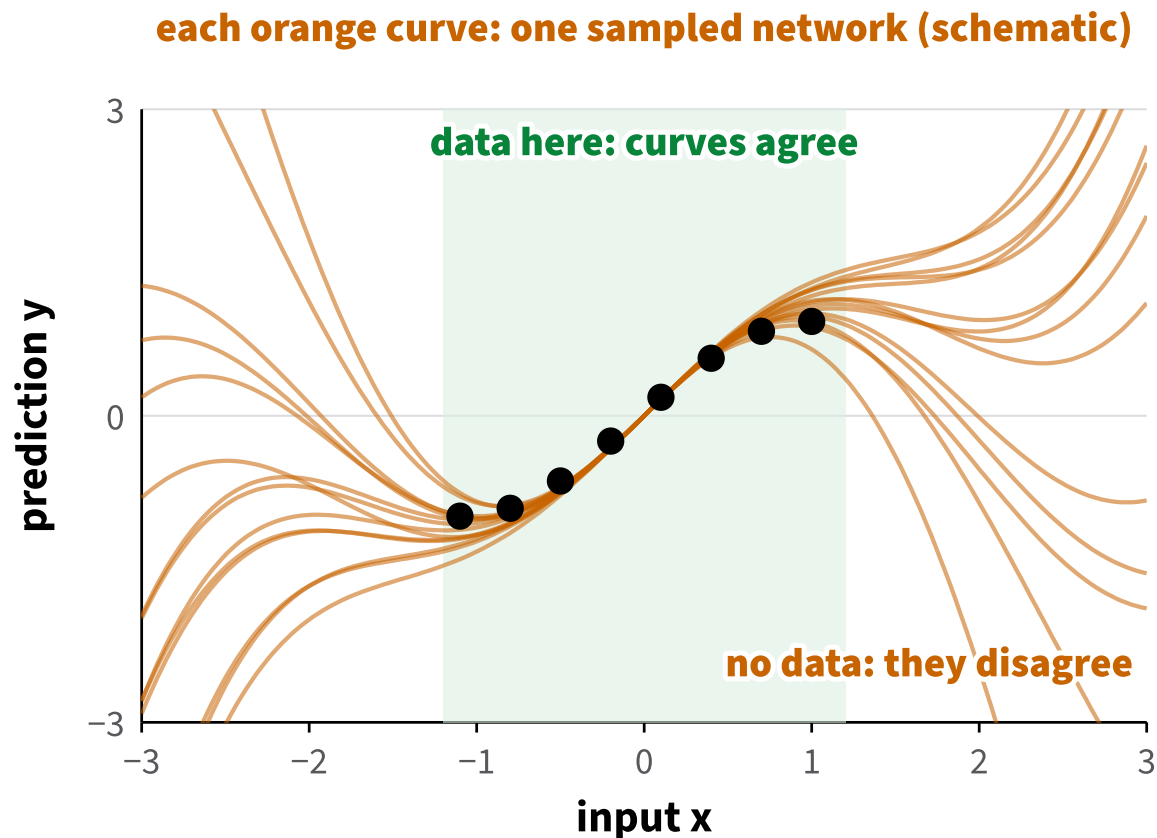
Use the **slope** from a **small random batch** plus a **random kick**, and skip the **accept test**.

USED FOR

Neural networks that can say how sure they are.

Trouble 3: too much data

Let the **walker** hop over a network's weights: the sampled networks show where they are unsure.



Q: Why not train just one network?

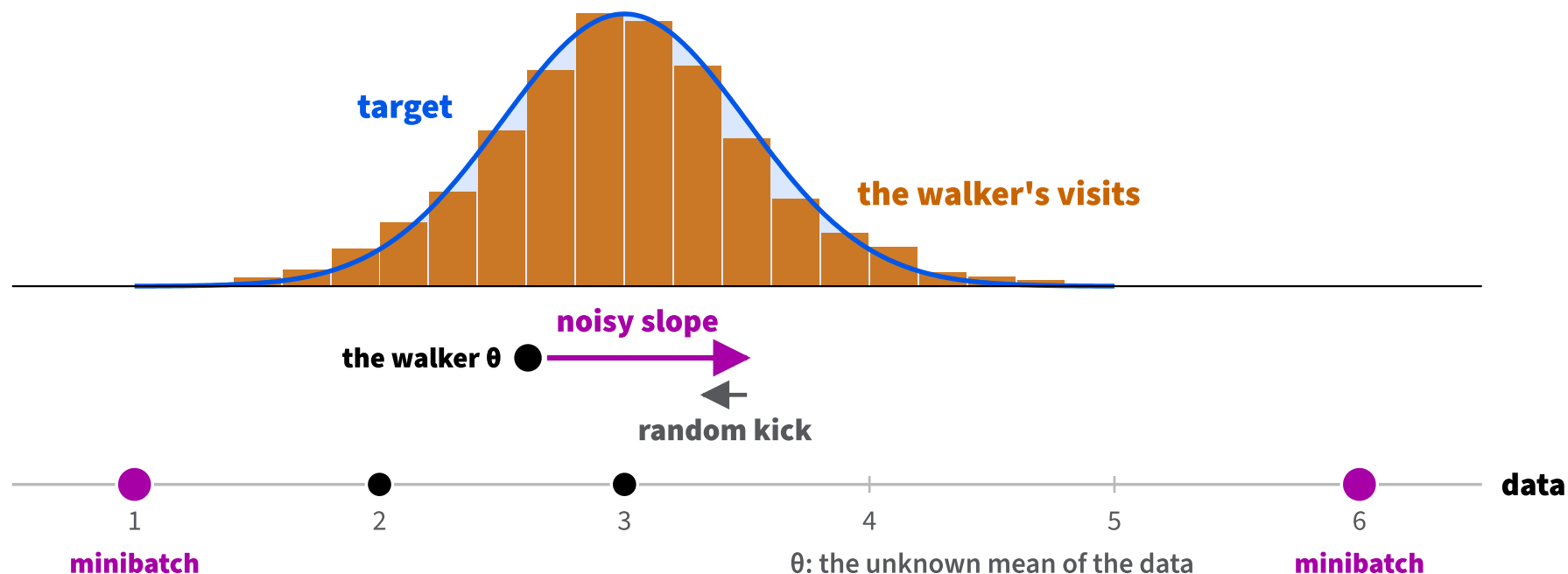
A: One network is one curve; it never says “not sure”.

Look: near the data the sampled networks agree; far from it they fan out.

Trouble: each **accept test** needs g , a product over every training example: **millions** of them.

Trouble 3: too much data

Fix: follow the **slope** from two random data points, add a **random kick**, skip the **accept test**.



Q: Isn't following the slope just optimisation? **A:** Without the kick, yes: it settles near the best θ . The kick spreads the visits like the target.

Try: switch the kick off. This is **stochastic-gradient Langevin dynamics** [Welling and Teh, 2011]; the slope of $\ln g$ never sees Z .

Outline

Sampling without Z (Lecture 4)

Troubles 1–3: measuring unknowns from data

► **Troubles 4–5: creating pictures and text**

Conclusion and your turn

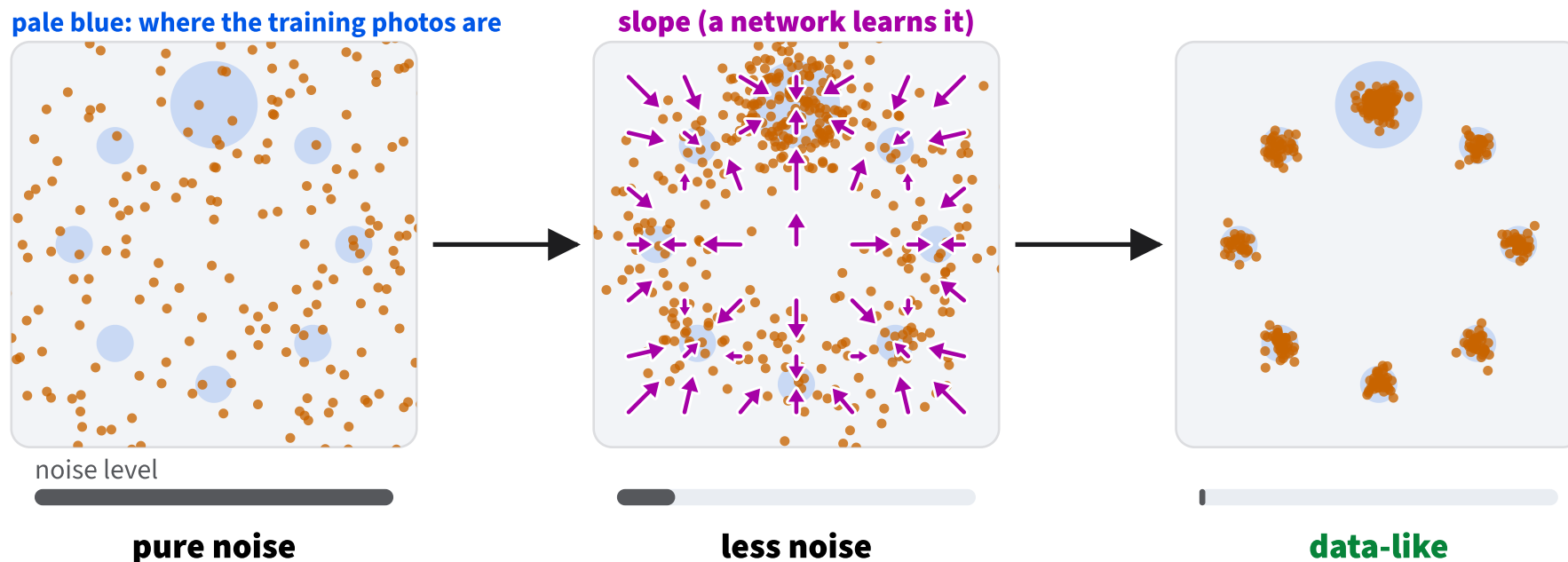
Next: each place the walker visits is a new picture or a new text: it creates.

Trouble 4 in three sentences

- BACKGROUND** For photos there is **no formula** for the **fit score**, only **millions** of example photos.
- KEY IDEA** Train a network to give the **slope** towards “more photo-like”, then **walk from pure noise** while the **noise shrinks**.
- USED FOR** Diffusion image generators, such as Stable Diffusion.

Trouble 4: no formula

Trouble: no formula for $g(\text{photo})$, only examples. **Fix:** learn the slope; walk from noise.



Watch: each dot is one whole picture (one coordinate per pixel; 2 shown). Heavy noise blurs the 8 clusters into one hill, like the **hot walker (p. 14)**; following the slope (**p. 21**) as the **noise shrinks**, the dots settle into all 8, about 40% on the big disk (which holds 44% of the photos).

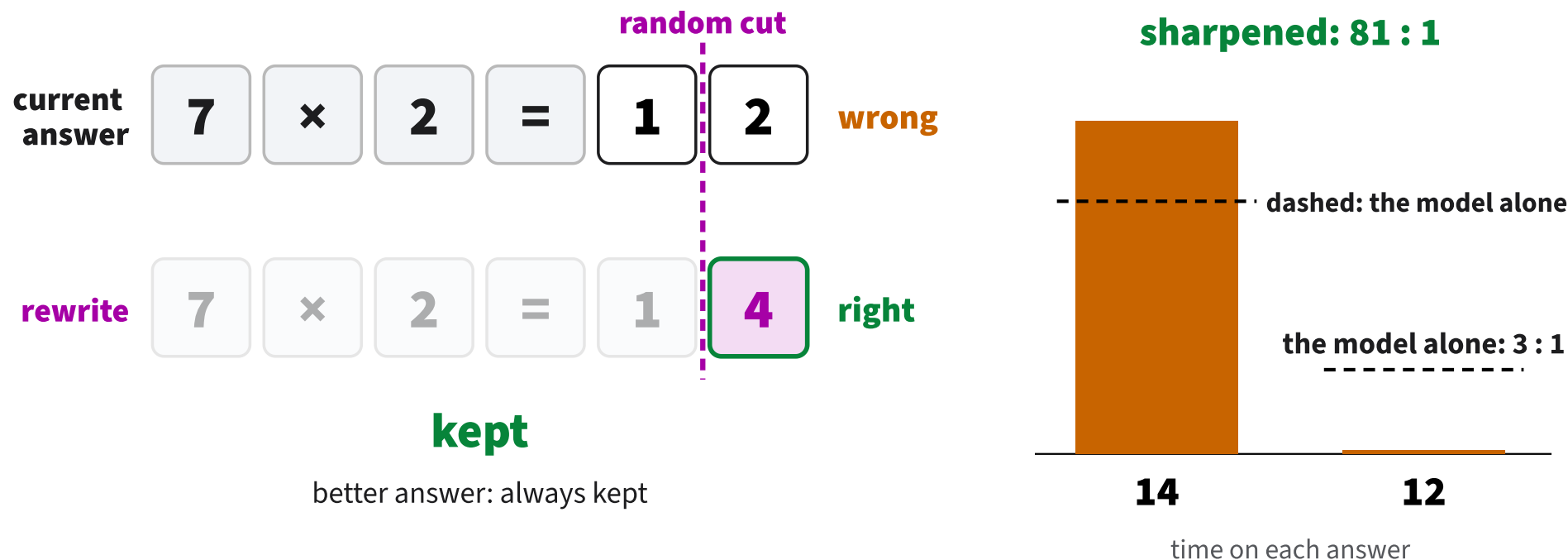
Try: start small: only 14% reach the big disk. Why? This is the idea behind **diffusion image generators** such as Stable Diffusion [Song and Ermon, 2019; Ho et al., 2020].

Trouble 5 in three sentences

- BACKGROUND** A language model gives each answer a probability; the right answer is often the most likely, but not likely enough.
- KEY IDEA** **Sharpen:** **sample** answers with probability raised to a power. The **walker** **rewrites the end** of an answer and keeps it by the **accept test**, so no new Z is needed.
- USED FOR** Better maths and coding answers from the same model, with no extra training (2025).

Trouble 5: a sum over all texts

Trouble: sharpening needs a new Z over all texts. **Fix:** rewrite the end; the accept test decides.



Q: Why not just take the most likely answer? **A:** The likeliest word at each step does not give the likeliest whole answer; finding that needs a search over all texts.

Watch: power 4 turns 3 : 1 into 81 : 1, with no new Z . **Try:** $\alpha = 1$ and $\alpha = 8$. This is **power sampling** [Karan and Du, 2025].

Outline

Sampling without Z (Lecture 4)

Troubles 1–3: measuring unknowns from data

Troubles 4–5: creating pictures and text

► **Conclusion and your turn**

Finally: the five fixes side by side, what is still open, and what you can try yourself.

Conclusion

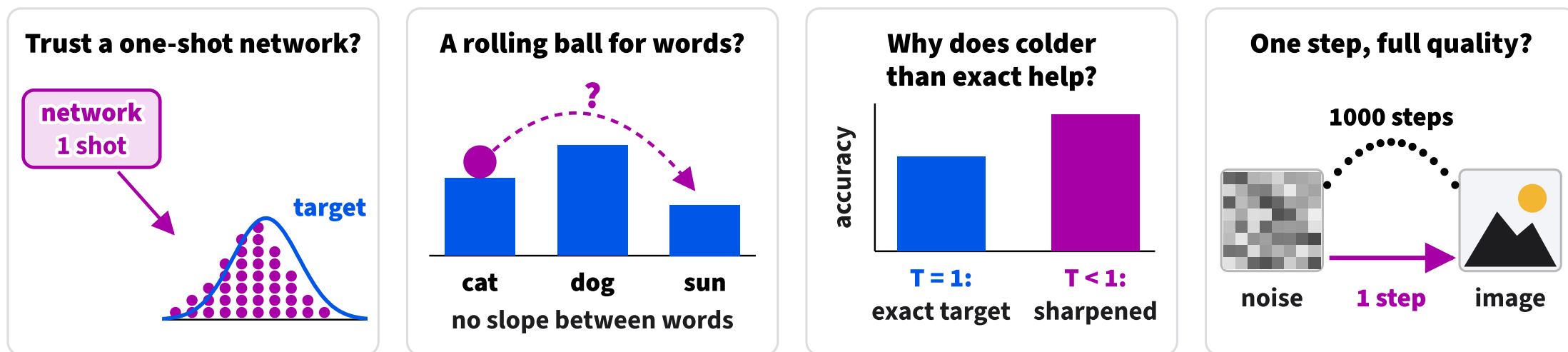
One **walker**, five small changes, and Z is never computed.

	Trouble	Fix	Used for
1	far-apart hills	a hot walker, swaps	colliding black holes
2	many unknowns at once	slide like a marble	COVID-19 lockdowns
3	too much data	minibatch slope + kick	neural-network error bars
4	no formula for g , only examples	learned slope, from noise	image generators
5	sharpening: a new Z over all texts	rewrite the end, test	language-model reasoning

Read: fix 1 adds a hot copy; fixes 2–5 change how the next **candidate** is proposed, and fixes 3–4 also drop the **accept test**. None computes Z : a ratio of **fit scores** cancels it, and a **slope** never sees it.

Conclusion

- ▶ It is possible to **sample without computing Z** : compare two candidates, or follow the **slope**.
- ▶ Each frontier adds a copy, changes the proposal, or drops the **accept test**.
- ▶ We leave four **open problems**:



Think: each would make a good course project; start from Dax et al. (2021) for 1, Wenzel et al. (2020) for 3 and Song et al. (2023) for 4.

Your turn

Predict first, then try: every animation in this lab has a knob.

In these slides

- ▶ the **fit score** of B (p. 6): where do the **walkers** settle?
- ▶ the distance between hills (p. 13): when does the walker cross?
- ▶ the temperature T (p. 14): is hotter always better?
- ▶ the roll length (p. 18) and the **random kick** (p. 21)
- ▶ the noise schedule (p. 24) and the power α (p. 26)

After class: `lab4_explore.py`

1. The walker: do the **visits** follow **fit scores** 1, 2, 5?
2. Far-apart hills: how fast does the wait grow?
3. A **hot walker**: write the **swap** test.
4. Generate from noise: why shrink it slowly?

Course site → Week 4 → Laboratory → Lab files. Each experiment needs one or two lines of Python.

Tip: write your prediction down before you click; the surprises are where the learning is.

Selected References I

Start here: Hastings (1970) explains the walker itself; the others are the methods and today's results.

-  [Abbott, B. P., et al. \(LIGO Scientific Collaboration and Virgo Collaboration\) \(2016\).](#)
Properties of the binary black hole merger GW150914.
Physical Review Letters, 116(24):241102.
-  [Dax, M., Green, S. R., Gair, J., Macke, J. H., Buonanno, A., and Schölkopf, B. \(2021\).](#)
Real-time gravitational wave science with neural posterior estimation.
Physical Review Letters, 127(24):241103.
-  [Duane, S., Kennedy, A. D., Pendleton, B. J., and Roweth, D. \(1987\).](#)
Hybrid Monte Carlo.
Physics Letters B, 195(2):216–222.
-  [Event Horizon Telescope Collaboration \(2019\).](#)
First M87 Event Horizon Telescope results. VI. The shadow and mass of the central black hole.
The Astrophysical Journal Letters, 875(1):L6.
-  [Flaxman, S., Mishra, S., Gandy, A., et al. \(2020\).](#)
Estimating the effects of non-pharmaceutical interventions on COVID-19 in Europe.
Nature, 584:257–261.
-  [Geyer, C. J. \(1991\).](#)
Markov chain Monte Carlo maximum likelihood.
In Computing Science and Statistics: Proceedings of the 23rd Symposium on the Interface, pages 156–163.

Selected References II

Good entry points: Song and Ermon (2019) for image generation; Karan and Du (2025) for language models.



Hastings, W. K. (1970).

Monte Carlo sampling methods using Markov chains and their applications.

Biometrika, 57(1):97–109.



Ho, J., Jain, A., and Abbeel, P. (2020).

Denoising diffusion probabilistic models.

In Advances in Neural Information Processing Systems 33, pages 6840–6851.



Karan, A. and Du, Y. (2025).

Reasoning with sampling: your base model is smarter than you think.

arXiv:2510.14901.



Metropolis, N., Rosenbluth, A. W., Rosenbluth, M. N., Teller, A. H., and Teller, E. (1953).

Equation of state calculations by fast computing machines.

The Journal of Chemical Physics, 21(6):1087–1092.



Song, Y. and Ermon, S. (2019).

Generative modeling by estimating gradients of the data distribution.

In Advances in Neural Information Processing Systems 32.

Selected References III

A short read: Welling and Teh (2011), eight pages on how sampling meets minibatch training.



Song, Y., Dhariwal, P., Chen, M., and Sutskever, I. (2023).

Consistency models.

In Proceedings of the 40th International Conference on Machine Learning, PMLR 202:32211–32252.



Swendsen, R. H. and Wang, J.-S. (1986).

Replica Monte Carlo simulation of spin-glasses.

Physical Review Letters, 57(21):2607–2609.



Veitch, J., Raymond, V., Farr, B., et al. (2015).

Parameter estimation for compact binaries with ground-based gravitational-wave observations using the LALInference software library.

Physical Review D, 91(4):042003.



Welling, M. and Teh, Y. W. (2011).

Bayesian learning via stochastic gradient Langevin dynamics.

In Proceedings of the 28th International Conference on Machine Learning, pages 681–688.



Wenzel, F., Roth, K., Veeling, B. S., Świątkowski, J., Tran, L., Mandt, S., Snoek, J., Salimans, T., Jenatton, R., and Nowozin, S. (2020).

How good is the Bayes posterior in deep neural networks really?

In Proceedings of the 37th International Conference on Machine Learning, PMLR 119:10248–10259.

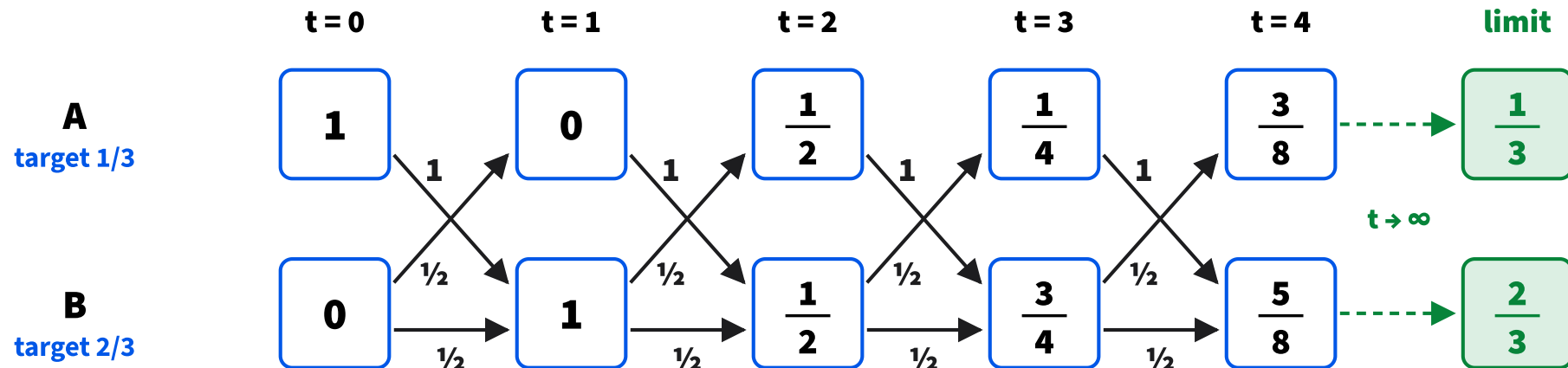
Thank you!

Next: Lab 5 (Saturday 10 October), the expectation–maximization algorithm at the frontier

Try the walker yourself: the ten-line Python version is on the backup slides that follow.

Backup: the walker of Lecture 4

Why the **walkers** settle at one third and two thirds: the flows balance.



Look: each box is the share of walkers at step t : the sum of (previous box $\times$ arrow). The shares stop changing when the flow $A \rightarrow B$, $\frac{1}{3} \cdot 1$, equals the flow $B \rightarrow A$, $\frac{2}{3} \cdot \frac{1}{2}$: detailed balance (Lecture 4, Theorem 9).

Backup: the walker of Lecture 4

The whole **walker** is a few lines of Python, and it never uses Z .

```
import random
random.seed(0)
g = {"A": 1, "B": 2}                                # fit scores; Z is never computed
x, visits = "A", {"A": 0, "B": 0}
for t in range(100_000):
    y = "B" if x == "A" else "A"                    # propose the other picture
    if random.random() < min(1, g[y] / g[x]):         # the accept test: better fit always, worse g(new)/g(old)
        x = y
    visits[x] += 1
print({s: v / 100_000 for s, v in visits.items()})
{'A': 0.33392, 'B': 0.66608}
```

Try: change the **fit scores** 1 and 2 to any two numbers; the shares follow their ratio, and Z is still never computed. More in `lab4_explore.py`.

Backup: Trouble 1

The **swap** of parallel tempering is again a comparison.

A copy at temperature T walks on $g(x)^{1/T}$. Two copies at $T_1 < T_2$ propose to swap their states x_1, x_2 ; the **accept test** compares the swapped pair with the current pair:

$$\frac{g(x_2)^{1/T_1} \cdot g(x_1)^{1/T_2}}{g(x_1)^{1/T_1} \cdot g(x_2)^{1/T_2}} = \left(\frac{g(x_2)}{g(x_1)} \right)^{\frac{1}{T_1} - \frac{1}{T_2}}.$$

Accept with $\min\{1, \text{this ratio}\}$. In the animation $T_1 = 1$ and $T_2 = 10$, so the power is $\frac{9}{10}$.

Notice: only the ratio $g(x_2)/g(x_1)$ appears, and the swap keeps the **walker**'s **visits** correct. Like every step, it never needs Z .

Backup: Trouble 2

One leapfrog step, by hand (a roll chains many of them).

Height $U(x) = -\ln g(x) = \frac{1}{2} \cdot x^2$, **slope** $U'(x) = x$. Start at $x = 1$ with **momentum** $v = 1$, step 0.5; energy $H = U(x) + \frac{1}{2} \cdot v^2 = 1$.

$$v = 1 - 0.25 \cdot 1 = 0.75$$

$$x = 1 + 0.5 \cdot 0.75 = 1.375$$

$$v = 0.75 - 0.25 \cdot 1.375 = 0.406$$

$$H' = \frac{1}{2} \cdot 1.375^2 + \frac{1}{2} \cdot 0.406^2 = 1.028$$

$$\min\{1, e^{-(1.028-1)}\} = 0.97$$

half kick: speed $- 0.25 \cdot$ slope

drift: position $+ 0.5 \cdot$ speed

half kick

energy almost kept

the **accept test**, with **fit score** e^{-H}

Notice: in this one step the energy moves only from 1 to 1.028, so the accept test keeps it 97% of the time; a roll is many such steps.